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Topos Theory and Categorical Logic: A Comprehensive Treatment

What is a Topos?

A **topos** is a category that behaves like the category of sets, but with more flexibility. Topoi provide: - An alternative foundation for mathematics (not ZFC) - Categorical treatment of geometry (via sheaves) - Constructive/intuitionistic mathematics naturally - Internal logic (automatic reasoning within the topos)

Elementary Topoi: Complete Axiomatization

Formal Definition

A category $\mathcal{E}$ is an **elementary topos** if:

1. Finite Limits: - Has terminal object 1 - Has binary products $X \times Y$ for all X, Y - Has equalizers for parallel pairs $f, g: X \rightarrow Y$

Consequence: Has all finite limits (products, pullbacks, etc.)

2. Cartesian Closure: - For all $X, Y \in \mathcal{E}$, there exists internal hom $[X, Y]$ - Exponential adjunction: $\text{Hom}(X \times Y, Z) \cong \text{Hom}(X, [Y, Z])$

3. Subobject Classifier: - Existence of object Ω with universal property: - Subobjects of $X \leftrightarrow$ Morphisms $X \rightarrow \Omega$ (bijection) - For each monic $m: S \rightarrow X$, unique χ_m with: $S \xrightarrow{m} X \xrightarrow{\chi_m} \Omega \xrightarrow{T} \Omega$

[pullback diagram]

Where $T: 1 \rightarrow \Omega$ is “true” element.

Why These Axioms?

Finite Limits

- **Terminal 1:** Single element (like $\emptyset$ or point)
- **Products:** Combine structures (like Cartesian product)
- **Equalizers:** Define subobjects by equations
- **Together:** Allow constructing all finite limits (universal properties)

Cartesian Closure

- **$[X, Y]$:** Internalize morphisms as objects
- **Exponential law:** Move arguments in/out of exponents
- **Allows:** Function spaces, higher-order structures
- **Consequence:** Topos is a **closed symmetric monoidal category** (with $\times$ as tensor)

Subobject Classifier Ω

- **Generalizes {true, false}:** Ω is “object of truth values”
- **Subobjects via morphisms:** Instead of “ $X \subseteq Y$ ”, use characteristic function
- **Makes logic internal:** Type theory works inside topos automatically

Examples of Elementary Topoi

1. Set: The Prototypical Topos

Objects: Sets
 Morphisms: Functions
 Limits: Ordinary set operations
 $\Omega = \{\text{true}, \text{false}\}$
 $[X, Y] = Y^X$ (functions $X \rightarrow Y$)
 Logic: Classical (every truth value is true or false)

2. Presheaf Topoi: $[C^{\text{op}}, \text{Set}]$

Objects: Functors $C^{\text{op}} \rightarrow \text{Set}$ (presheaves)
 Morphisms: Natural transformations
 Limits: Pointwise (computed in Set)

Example: Simplicial sets = $[\Delta^{\text{op}}, \text{Set}]$

Key property: Always a topos!

3. Sheaf Topoi: $\text{Sh}(X)$

Objects: Sheaves on topological space X
 Morphisms: Sheaf morphisms
 Limits: Pointwise

Subobject classifier Ω :
 – $\Omega(U) = \{\text{open sets } V \subseteq U\}$ (all opens)
 – Open sets glue by unions

Characteristic function: $\chi_S(U) = \{V \subseteq U \mid V \subseteq S\}$

4. Effective Topos: Eff

Objects: Assemblies (sets with realizability notion)
 Morphisms: Realizable functions
 Logic: Intuitionistic (excludes law of excluded middle)
 Church's thesis: true as internal axiom
 Application: Computability theory

5. G-Sets: G-Equivariant Sets

Objects: Sets with G -action (G finite group)
 Morphisms: G -equivariant functions

$\Omega = G$ -set of subgroups (with action)
 Classifier: $\chi_S(g) = \text{stability group of } gS$

6. Simplicial Sets: $\text{sSet} = [\Delta^{\text{op}}, \text{Set}]$

Δ = simplicial category (finite non-empty ordinals)
 Objects: X_n = n -simplices (with face/degeneracy)
 Morphisms: Simplicial maps

Logic: Classical (Set-based)
 Application: Homotopy theory, ∞ -categories

7. Realizability Topos: $\text{RT}(A)$

A = partial combinatory algebra (computation model)
 Objects: Assemblies (sets with realizable proofs)
 Morphisms: Realizable functions

Church-Turing thesis true internally
 Application: Recursive mathematics

Subobject Classifier and Logical Structure

The Power Object Ω

Definition: Ω is the **terminal object** in the category of subobject classifiers.

Universal property:

For any monic $m: S \rightarrow X$:

$\exists! \chi_m$ (characteristic function) making diagram pullback:

$$\begin{array}{ccc} S & \xrightarrow{m} & X \\ | & & | \\ | & & \chi_m \\ | & & | \\ 1 & \xrightarrow{\tau} & \Omega \end{array}$$

Elements of Ω : Truth values in the topos

In Set: $\Omega = \{T, F\}$ In $\text{Sh}(X)$: $\Omega(U) = \text{opens of } U$ In Eff: $\Omega = \text{partial recursive functions}$

Power Objects: Ω^X

Definition: $P(X) = [X, \Omega]$ (exponential in topos)

Universal property: Morphisms $X \rightarrow \Omega$ correspond to subobjects of X

Comprehension:

$\{x \in X \mid \varphi(x)\} \leftrightarrow \text{characteristic function } X \rightarrow \Omega$

Higher-order logic: $P(P(X))$ exists, giving impredicative quantification

Grothendieck Topoi and Sheaves

Sites and Grothendieck Topologies

A **Grothendieck topology** on small category C : - For each $X \in C$, collection $\text{Cov}(X)$ of “covering families” - Must satisfy: 1. Identity covers (single id morphism) 2. Stability under pullback 3. Transitivity (covers of covers)

Examples: - Zariski topology (on schemes): Open covers in $\text{spec}(R)$ - Étale topology: Étale morphisms as covers - Canonical: All families with epi legs (arrows)

Sheaves on Sites

Presheaf $F: C^{\text{op}} \rightarrow \text{Set}$

Sheaf condition: For covering family $\{U_i \rightarrow X\}$:

$F(X) \rightarrow \prod F(U_i) \Rightarrow \prod F(U_i \times_X U_j)$
(matching condition: gluing axiom)

Consequence: F is **continuous** with respect to Grothendieck topology

Sheafification

Theorem: Every presheaf extends to sheaf (sheafification functor)

Presheaves $\supseteq$ Sheaves
(lex reflective subcategory)

Construction: $F^+ =$ sheafification - $F^+(X) =$ colimit of compatible families of F -sections - Natural map $F \rightarrow F^+$ universal

Topoi as Foundations for Mathematics

Why Not ZFC?

ZFC Axioms: - Materialism (elements matter) - Global membership $\in$ - Law of excluded middle - Axiom of choice

Alternative in topoi: - Structuralism (relationships matter) - Local logic internal to topos - Constructive/intuitionistic reasoning - Choice axiom optional

Elementary Topos Category Theory (ETCS)

Lawvere-MacLane: Axioms for category of sets (category-theoretic)

Instead of axioms on $\in$, axioms on: - Objects (sets) - Morphisms (functions) - Composition, identity - Finite limits, colimits, exponentials

Theorem (Lawvere): ETCS equivalent to bounded ZFC

Natural Numbers Objects (NNO)

Definition: N is NNO if:

$0: 1 \rightarrow N$
 $s: N \rightarrow N$ (successor)

$\forall f: X \rightarrow X$ and $e: 1 \rightarrow X$, $\exists! u: N \rightarrow X$:
 $u \circ 0 = e$
 $u \circ s = f \circ u$
(recursion principle)

Consequence: Can define all arithmetic in topos!

Internal Logic of Topoi

Mitchell-Bénabou Language

Internal language: Reason “as if” objects are sets

Kripke-Joyal Semantics: Truth notion for internal formulas

For formula $\phi(x)$ and stage U :

$U \models \phi(x)$ iff characteristic function lands in true

More precisely: $U \rightarrow [1, \Omega]$ lands in $T: 1 \rightarrow \Omega$

Intuitionistic Logic

In topoi (especially non-Boolean): - **Law of excluded middle fails:** $A \vee \neg A$ not provable - **Double negation:** $\neg\neg A$ not equivalent to A - **Proof by contradiction:** Not valid in constructive logic - **BHK interpretation:** Proofs are constructions, not just existence claims

Examples of Intuitionistic Reasoning

In effective topos:

$\forall x \exists y \phi(x, y)$ [universal + existential]

$\neq \exists f \forall x \phi(x, f(x))$ [requires choice function!]

Church's thesis: all functions are computable

Geometric Morphisms

Definition

Geometric morphism $f: \mathcal{E} \rightarrow \mathcal{F}$ ($f: \mathcal{F} \rightarrow \mathcal{E}$ in opposite direction):

$f^*: \mathcal{F} \rightarrow \mathcal{E}$ (inverse image, must be left exact)

$f_*: \mathcal{E} \rightarrow \mathcal{F}$ (direct image, right adjoint to f^*)

Left exactness of f^* : Preserves finite limits

Key property: $f^* \dashv f_*$ always holds!

Essential Geometric Morphisms

When f_* also has left adjoint $f^!$:

$f^! \dashv f^* \dashv f_*$
(essential morphism)

Consequence: Much stronger, allows comparing logic

Boolean Topoi

Topos $\mathcal{E}$ is Boolean if:

$\Omega = \{T, F\}$ (two-element set)

Equivalently: Law of excluded middle
 Equivalently: Every object decidable

Theorem (Barr): Every topos has Boolean sheafification

Applications Throughout Mathematics

1. Sheaf Cohomology

$$H^i(X, F) = \text{Ext}^i(\mathcal{O}_X, F)$$

Define via exact sequences in $\text{Sh}(X)$
 Works for any sheaf on any topos

2. Étale Topology and Algebraic Geometry

Scheme X has étale topology
 Sheaves measure étale structure
 Adic rings from sheaf sections

3. Effective Topos and Computability

Every morphism is algorithm
 Decidability is propositional form
 Computability provable internally

4. Synthetic Differential Geometry

Smooth topos SDG:
 – Infinitesimals exist (dx)
 – Taylor expansion automatic
 – Derivatives from logic

5. Homotopy Type Theory

Types are ∞ -groupoids
 Paths are proofs of equality
 Univalence: equivalence = equality

∞ -topoi (Lurie): Higher sheaf theory

Comparisons: Topoi vs Set Theory

Aspect	ZFC	Topos
Foundation	Sets + membership	Categories + morphisms
Logic	Classical (LEM)	Constructive (default)
Geometry	External	Internal + via sheaves
Functions	Global	Local (sheaves)
Truth values	$\{T, F\}$	Ω object (variable)
Choice	Axiom	Optional

Common Mistakes

- ✗ **Mistake:** “Topos = generalization of Set” ✓ **Reality:** Topos = flexible foundation, Set is special case
- ✗ **Mistake:** “Topoi must satisfy classical logic” ✓ **Reality:** Default is intuitionistic; Boolean is special
- ✗ **Mistake:** “Sheaves are only topological” ✓ **Reality:** Sheaves work on any site (Zariski, étale, etc.)
- ✗ **Mistake:** “Subobject classifier is just truth values” ✓ **Reality:** Ω is sophisticated object controlling all logic

Study Guide

1. Master finite limits and cartesian closure
2. Understand subobject classifier deeply
3. Work through presheaf example $[C^{op}, Set]$
4. Study sheafification on topological space
5. Learn internal logic (Mitchell-Bénabou)
6. Explore Kripke-Joyal semantics
7. Apply to effective topos
8. Connect to homotopy type theory

Further Reading

- Mac Lane & Moerdijk, “Sheaves in Geometry and Logic” (foundational)
- Johnstone, “Sketches of an Elephant” (comprehensive reference)
- Taylor, “Practical Foundations of Mathematics” (constructive approach)
- Univalent Foundations, “Homotopy Type Theory” (modern direction)

Research Questions

1. Why must subobject classifier exist?
2. How does logic emerge from Ω ?
3. Why is classical logic exceptional?
4. What makes Grothendieck topologies fundamental?
5. How do topoi generalize ZFC?
6. Why is sheafification universal?